

Arithmetic Transseries

Beyond an Accumulation Cut

Divisibility, summable inversion, and
curvature-lifted resonances

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Abstract

Divisor-counting sequences have corrections with arithmetic coefficients $\mathbf{1}_{d|n}$ and exponential actions $a(1 - 1/d)$ accumulating at a . This combination defeats finite-grid truncation without defeating Hahn well-ordering. We construct an arithmetic Hahn ring and a compatible scale of weighted Banach algebras that retain the indicators exactly. An lcm convolution handles arithmetic multiplication, while an action-label weight makes the entire accumulating block summable on expanding families of analytic sheets. For the elementary input block, membership has the exact criterion $\sum_d |c_d|e^{-ad} < \infty$.

We prove a convergent, all-action, fixed-label inverse theorem for $e^{ax}(1 + E_n(x))/x$, with explicit coefficients and a geometric remainder. It applies to necklaces and primitive necklaces on their exact ranges. A second result identifies the first inverse sector beyond the primitive action cut: it has action a , but its leading pure-exponential amplitude cancels, leaving a curvature-induced factor of order X^{-1} . An exact coefficient transformation explains this phenomenon at every homogeneous order and gives its extension to cores $e^{ax}x^{-b}$.

The construction answers a specified algebra-and-summation direction in the ProveIt companion volume. It is not a canonical interpolation of an arithmetic sequence, a field containing nontrivial indicators, or a general resurgence theorem. Full mathematical proofs are supplied; 964 exact finite assertions and 48 high-precision error checks provide supplementary audits. These results are not Lean-verified, and publication-level priority has not been established.

Status. AI-assisted research manuscript; unrefereed. Classical ingredients and proposed extensions are distinguished in Section 1. The repository snapshot, reproducibility instructions, and formalization boundaries are recorded explicitly.

Contents

1	The research target and the scope of the answer	2
1.1	A specific question in the repository	2
1.2	What is established	2
1.3	Classical ingredients and proposed extensions	3
2	Divisor sectors and the two variables	3
2.1	The exact combinatorial input	3
2.2	Why the accumulation point matters	4
3	The arithmetic Hahn ring	4
3.1	The action monoid	4
3.2	Divisor coordinates and lcm multiplication	5
3.3	The formal completion	5
4	An action-label Banach completion	6
4.1	An exact admission criterion	7
4.2	Functional calculus, derivatives, and translation	8
5	All-action inversion on arithmetic sheets	8
5.1	The exact core and the reversion equation	8
5.2	The algebraic reversion identity	9
5.3	A quantitative convergence theorem	10
5.4	An exact inverse on the range	11
6	Coefficients, collisions, and the first post-cut sector	12
6.1	The first three homogeneous kernels	12
6.2	The arithmetic content of action collisions	12
6.3	Cancellation at the accumulation boundary	13
7	An all-order resonance law	14
7.1	The pure-exponential comparison	15
7.2	General power-law factors	15
8	Obstructions to stronger interpretations	16
8.1	Indicators cannot live faithfully in a field	16
8.2	A compact profinite parameter does not repair summation	16
8.3	Bounded-action extraction versus homogeneous truncation	17
9	Finite algorithms and reproducible audits	17
9.1	Exact membership and a fixed-action coefficient algorithm	17
9.2	Efficient homogeneous jets for a known label	18
9.3	What the accompanying program checks	18
10	Consequences and interpretation	19
10.1	A usable answer at an infinite action boundary	19
10.2	A precise beyond-all-orders effect	20
10.3	What remains outside the theorem	20
11	Further research questions	20
12	Conclusion	22
A	Repository provenance and scope ledger	22
B	Lean interfaces and a proposed verification route	23
C	Reproducibility and deliverable contents	23

1 The research target and the scope of the answer

1.1 A specific question in the repository

The current ProveIt transseries project separates its generic Lean modules from the Fabius-function development. Its companion volume, *Combinatorial Transseries and Their Inverses*, contains a chapter on necklaces, Lyndon words, and irreducible polynomials. The subsequent “Further research directions” section explicitly asks for an arithmetic transseries algebra that preserves divisibility indicators, permits an accumulation of actions at $\log q$, and specifies summable products and inverse operations [1]. This is the target of this article.

The source consulted was the file named in Appendix A, at the pinned commit `db68f0853c3c` (full hash in the appendix). In that source the question occurs near line 4971. Its benchmark is Proposition `t2:prop:necklace-cutoff`; the relevant chapter has label `ct:ch:necklaces`. The source already provides finite divisor cutoffs, finite-cutoff trigonometric interpolations, certain fixed-radical analytic sheets, and integer threshold methods. None of those constructions is claimed here as a new result.

The repository also contains an unmerged article on action accumulation and nonlinear inversion using Laplace measures. Its own catalogue entry states that it does not preserve the divisibility indicators and therefore does not provide the arithmetic algebra requested by the companion [2]. The present construction addresses that missing arithmetic component rather than repeating the measure-valued theory.

1.2 What is established

There are three main results, together with their necessary limitations.

Arithmetic summation across the cut. Theorems 4.1 and 4.3 provide an explicit weighted Banach algebra whose coefficients multiply by lcm convolution. The whole infinite family of primitive actions is retained. Scalar analytic functional calculus, including logarithms and multiplicative inverses near one, is norm-convergent. Evaluation gives exact arithmetic data on the diagonal $x = n$ and holomorphic functions on label-dependent half-planes.

Convergent inverse transport. Theorems 5.2 and 5.5 give an explicit inverse series around the exact core e^{aX}/X . The series is absolutely summable even after it is expanded into individual action tuples. A uniform bound controls its tail. The arithmetic label is fixed throughout an inverse calculation; this is an essential part of the statement.

Curvature-lifted resonance. Theorems 6.1 and 7.1 identify a cancellation at the first action beyond the primitive block and explain it at every homogeneous order. For the more general core $e^{ax}x^{-b}$, Corollary 7.2 gives the first surviving coefficient at each pure-exponential resonance.

The first two results supply a constructive answer to the formal, analytic-functional-calculus, and fixed-label reversion portions of the repository question. They deliberately do not postulate a fractional shift of the arithmetic label. Section 8 proves why ordinary differential-field and compact-profinite interpretations cannot simply be imposed on these data.

1.3 Classical ingredients and proposed extensions

The multiplication $\sum_{\text{lcm}(i,j)=l} c_i b_j$ is the classical lcm convolution. Its relationship with the divisor-zeta transform is recorded, for example, in Tóth's survey, Section 4.4 [4]. It is not introduced here as a new arithmetic operation. Hahn supports, well-ordering arguments, and Lagrange inversion are also classical ingredients; Edgar's account gives background on transseries and their usual differential and compositional operations [5].

The proposed contributions are the combined action–label completion, its exact elementary-block admission criterion and faithful sheet realization, the absolute inverse transport through the accumulating arithmetic support, and the all-order curvature calculation and its post-cut consequence. Each has a proof below. A targeted literature check also found a 2026 preprint on analytic properties of necklace polynomials; it studies the alphabet-variable polynomial and its analytic inequalities, rather than the length-variable construction considered here [6]. This comparison is not an exhaustive novelty search. The results should be regarded as proposed research extensions, not as a verified claim of first discovery.

2 Divisor sectors and the two variables

Throughout, $\mathbb{N} = \{1, 2, \dots\}$ and $a > 0$. For integer alphabet size $q \geq 2$ we specialize to $a = \log q$. Set

$$e_d(n) = \mathbf{1}_{d|n}, \quad \lambda_d = a \left(1 - \frac{1}{d}\right), \quad d \geq 2. \quad (2.1)$$

The letter n always denotes an arithmetic label. The variable x is a real or complex analytic coordinate. They can be set equal, but they are not identified while differentiating or reverting a function.

2.1 The exact combinatorial input

The number of necklaces and primitive necklaces of length n are, respectively,

$$N_q(n) = \frac{1}{n} \sum_{d|n} \varphi(d) q^{n/d}, \quad L_q(n) = \frac{1}{n} \sum_{d|n} \mu(d) q^{n/d}. \quad (2.2)$$

For completeness, the first formula follows by averaging the fixed words of the cyclic rotations. A rotation through r positions fixes $q^{\gcd(n,r)}$ words, and grouping the rotations by $d = n / \gcd(n, r)$ gives $\varphi(d)$ copies. For the second formula, let $P_q(n)$ count words of least period n . Every word has a unique least period dividing n , so $q^n = \sum_{d|n} P_q(d)$. Möbius inversion then gives the numerator of the second expression, and every primitive orbit has n words. These are the same exact inputs used in the repository chapter [1].

Given coefficients c_d , define

$$E_n(x) = \sum_{\substack{d|n \\ d \geq 2}} c_d e^{-\lambda_d x}, \quad f_n(x) = \frac{e^{ax}}{x} (1 + E_n(x)). \quad (2.3)$$

Each E_n is an entire function because it is a finite sum. Choosing $c_d = \varphi(d)$ or $c_d = \mu(d)$ gives $f_n(n) = N_q(n)$ or $L_q(n)$. The family f_n is not asserted to be the restriction of a single interpolation of the counting sequence.

Two bounds used repeatedly below are

$$\sum_{d|n} \varphi(d) = n, \quad \sum_{d|n} |\mu(d)| \leq n. \quad (2.4)$$

The first identity counts the integers in $\{1, \dots, n\}$ by the order of the corresponding element of the cyclic group. The second follows because each summand is at most one and n has at most n divisors. Consequently, for either of the two counting inputs and real $X > 0$,

$$A_n(X) := \sum_{\substack{d|n \\ d \geq 2}} |c_d| e^{-\lambda_d X} \leq n e^{-aX/2}. \quad (2.5)$$

2.2 Why the accumulation point matters

The actions obey

$$\frac{a}{2} = \lambda_2 < \lambda_3 < \dots < a, \quad \lambda_d \longrightarrow a. \quad (2.6)$$

A finite additive grid with strictly positive generators has only finitely many elements below any fixed bound. The set in (2.6) does not. Moreover $2\lambda_2 = a$, so infinitely many primitive sectors precede the first quadratic sector.

This is not a failure of well-ordering: an increasing sequence can be well-ordered while accumulating at its supremum. The distinction between *finitely many decompositions of one action* and *finitely many actions below a bound* will be central. The former is true here; the latter is false.

For a finite divisor cutoff D , (2.4) gives

$$\left| E_n(n) - \sum_{2 \leq d \leq D} c_d e_d(n) e^{-\lambda_d n} \right| \leq n e^{-\lambda_{D+1} n}. \quad (2.7)$$

This recovers the companion's benchmark without changing its scope. The goal is now to sum the full primitive block and transport it through nonlinear operations, not to pretend that (2.7) resolves every bounded action with a fixed finite list.

3 The arithmetic Hahn ring

3.1 The action monoid

Let

$$S = \{ \lambda_{d_1} + \dots + \lambda_{d_m} : m \geq 0, d_i \geq 2 \}, \quad (3.1)$$

where the empty sum is zero. We order S by its usual real order.

Lemma 3.1 (Finite action decompositions). *The set S is well-ordered. Every $s \in S$ has only finitely many ordered representations by generators λ_d , including the length of the representation. Every fixed action therefore has finitely many decompositions into any prescribed finite number of elements of S .*

Proof. Every representation of an action at most B has length at most $\lfloor 2B/a \rfloor$. For a fixed length m , use the elementary well-quasi-order property of $\mathbb{N}^m$: every infinite sequence of tuples has an earlier tuple that is componentwise at most a later one. One proof is to successively extract an infinite nondecreasing subsequence in each coordinate. For a single coordinate, either a value occurs infinitely often or one can successively choose strictly increasing values.

An infinite strictly decreasing sequence of actions in S is bounded above by its first term. After selecting a subsequence, its chosen representations have a common length m . A componentwise comparable pair of tuples contradicts the strict decrease, because $d \mapsto \lambda_d$ is

strictly increasing. Thus there is no such decreasing sequence. A countable linearly ordered subset of $\mathbb{R}$ with no infinite strict descent is well-ordered: from a nonempty subset without a least element one could choose a strict descent recursively.

For a fixed s and fixed m , infinitely many distinct representing tuples would again yield a comparable pair. At least one coordinate would be strictly larger, which would make the sums unequal. There are only finitely many possible lengths. Finally, a decomposition into m elements of S can be refined into a generator representation, with marked cuts between its pieces. The generator representations and the possible cuts are finite; zero pieces cause only finitely many additional placements when m is prescribed. $\square$

Remark 3.2. This is the concrete positive-monoid argument required for the countably generated support. The repository's Neumann bridge supplies two-set product and finite-factorization statements, not by itself this entire countable monoid construction; see Appendix B.

3.2 Divisor coordinates and lcm multiplication

Let $\mathcal{A} = \mathbb{C}^{\mathbb{N}}$, with pointwise addition and multiplication. Every $A \in \mathcal{A}$ has unique divisor coordinates

$$A(n) = \sum_{l|n} c_l, \quad c_l = \sum_{d|l} \mu(l/d) A(d). \quad (3.2)$$

These expressions are finite for each index. Equivalently, $A = \sum_l c_l e_l$, where this notation means divisor-finite evaluation, not a claim of uniform convergence in n .

Proposition 3.3 (Classical lcm product). *If $A = \sum_i c_i e_i$ and $B = \sum_j b_j e_j$, then*

$$AB = \sum_l (c \star b)_l e_l, \quad (c \star b)_l = \sum_{\text{lcm}(i,j)=l} c_i b_j. \quad (3.3)$$

The inner sum is finite. The divisor-zeta map (3.2) is a ring isomorphism between this lcm-coordinate ring and $\mathcal{A}$.

Proof. The identity $e_i(n)e_j(n) = e_{\text{lcm}(i,j)}(n)$ follows directly from divisibility. If $\text{lcm}(i,j) = l$, then both i and j divide l , so there are only finitely many such pairs. Evaluating the right side at n gives the product of the two finite divisor sums. Invertibility of the coordinate map is Möbius inversion, proved by $\sum_{d|m} \mu(d) = \mathbf{1}_{m=1}$. $\square$

The construction in Proposition 3.3 is classical [4]; its role here is to preserve the arithmetic information at action collisions.

3.3 The formal completion

Definition 3.4. The arithmetic Hahn ring on S is

$$\mathcal{H} = \left\{ F = \sum_{s \in S} A_s t^s : A_s \in \mathcal{A} \right\}. \quad (3.4)$$

In divisor coordinates we write $F = \sum_{s,l} c_{s,l} e_l t^s$. Multiplication adds actions and takes lcms of labels:

$$(FG)_{u,l} = \sum_{\substack{s+t=u \\ \text{lcm}(i,j)=l}} c_{s,i} b_{t,j}. \quad (3.5)$$

Lemma 3.1 and the finite number of divisors of l make (3.5) a finite sum. Associativity follows by grouping finite triple decompositions; the identity is $e_1 t^0$. For any F with zero action-zero coefficient, every power F^m has actions at least $ma/2$. Hence a scalar power series $\sum_m a_m F^m$ is formally meaningful, since every exact action receives only finitely many contributions. In particular $\exp F$, $\log(1 + F)$, and $(1 + F)^{-1}$ are defined formally.

The universal arithmetic input is

$$E = \sum_{d \geq 2} c_d e_d t^{\lambda_d}. \quad (3.6)$$

It is legitimate for every coefficient sequence c_d at the formal level. Formal legitimacy alone does not give an analytic sum uniform in the label. The next section supplies the missing summation condition.

4 An action–label Banach completion

For a positive integer N and a real r with $0 \leq r < N$, put

$$h_N(l) = \max(N, l), \quad w_{N,r}(s, l) = e^{-s(h_N(l)-r)}, \quad \|F\|_{N,r} = \sum_{s \in S} \sum_{l \geq 1} |c_{s,l}| w_{N,r}(s, l). \quad (4.1)$$

Let $\mathcal{B}_{N,r}$ be the coefficient arrays of finite norm. The label factor is essential: a divisor l can occur on the diagonal only at integers at least l , and its exponential decay should be measured at that scale.

Theorem 4.1 (Arithmetic Banach algebra and sheet realization). *The space $\mathcal{B}_{N,r}$, with multiplication (3.5), is a commutative unital Banach algebra, and*

$$\|FG\|_{N,r} \leq \|F\|_{N,r} \|G\|_{N,r}. \quad (4.2)$$

For each $n \in \mathbb{N}$, the evaluation

$$\mathcal{E}_n F(x) = \sum_{s \in S} \sum_{l|n} c_{s,l} e^{-sx} \quad (4.3)$$

converges absolutely and is bounded by $\|F\|_{N,r}$ on $\Re x \geq \max(N, n) - r$. It is holomorphic in the interior, and $\mathcal{E}_n(FG) = \mathcal{E}_n(F)\mathcal{E}_n(G)$. The family of all sheet evaluations, with all $n \geq 1$, is faithful.

Proof. Weighted ℓ^1 is complete because multiplication of each coordinate by its positive weight is an isometric linear bijection to ordinary $\ell^1(S \times \mathbb{N})$. If $l = \text{lcm}(i, j)$, then $h_N(l) \geq h_N(i)$ and $h_N(l) \geq h_N(j)$. Since $s, t \geq 0$,

$$w_{N,r}(s+t, l) \leq w_{N,r}(s, i) w_{N,r}(t, j).$$

Apply the triangle inequality to (3.5), then sum nonnegative terms in either order. This proves (4.2). The identity has norm one.

If $l \mid n$, then $h_N(l) \leq \max(N, n)$. Throughout the stated half-plane, $|e^{-sx}| \leq w_{N,r}(s, l)$. The norm therefore majorizes the evaluation term by term. The same majorant proves uniform absolute convergence, including on compact subsets of the open half-plane, so the sum is holomorphic. Absolute convergence permits regrouping the product by actions and lcms, giving multiplicativity.

For faithfulness, suppose every sheet function is identically zero. Fix n and let $A_s(n) = \sum_{l|n} c_{s,l}$. If any such coefficient is nonzero, choose its least action s_0 , using Lemma 3.1. For large positive x ,

$$e^{s_0 x} \mathcal{E}_n F(x) = A_{s_0}(n) + \sum_{s > s_0} A_s(n) e^{-(s-s_0)x}.$$

The latter sum tends to zero by dominated convergence, using absolute convergence at one fixed sufficiently large x_0 . Thus the identically zero left side would tend to the nonzero number $A_{s_0}(n)$, a contradiction. All $A_s(n)$ vanish. Apply Möbius inversion for each s and all n to obtain $c_{s,l} = 0$. $\square$

Remark 4.2 (Faithfulness is not a tail-sequence assertion). The complete family of analytic sheets is used in the theorem. The nonzero arithmetic coefficient array $c_l = \mu(l)$ at action zero has evaluation $\mathbf{1}_{n=1}$. It vanishes on every diagonal integer $n \geq 2$. Thus one cannot infer formal equality merely from equality of eventual integer sequences in this unrestricted coefficient ring.

4.1 An exact admission criterion

Theorem 4.3 (Elementary-block admission). *For the block (3.6), the following are equivalent:*

$$E \in \mathcal{B}_{N,r} \text{ for one admissible pair } (N, r), \quad \sum_{d \geq 2} |c_d| e^{-ad} < \infty. \quad (4.4)$$

If these hold, then $E \in \mathcal{B}_{N,r}$ for every admissible pair, and for each fixed $r \geq 0$,

$$\|E\|_{N,r} \rightarrow 0 \quad (N \rightarrow \infty). \quad (4.5)$$

In particular, every polynomial-growth sequence c_d is admitted.

Proof. The part with $d \leq N$ is finite. For $d > N$ the norm summand is

$$|c_d| e^{-\lambda_d(d-r)} = |c_d| e^{-ad} e^{a+ar-ar/d}. \quad (4.6)$$

The last factor is bounded above and below by positive constants independent of d . This proves the equivalence and its independence of N, r . For fixed r , each fixed- d summand tends to zero as N grows, and it is bounded by $e^{ar} |c_d| e^{-a(d-1)}$. The latter sequence is summable under (4.4); dominated convergence proves (4.5). Polynomial growth is dominated by any positive exponential decay. $\square$

Remark 4.4. This is an exact criterion for membership in the specified absolute completion, not a necessary condition for the existence of each individual E_n : every such sheet is a finite sum regardless of the growth of c_d . The additional content is uniform control over the expanding sheet family and its nonlinear operations.

For $|c_d| \leq Cd$ one has a convenient explicit majorant. Writing $z = e^{-a}$, split the norm at $d = N$ and use $\lambda_d \geq a/2$ below the split and (4.6) above it:

$$\|E\|_{N,r} \leq \frac{CN(N+1)}{2} e^{-a(N-r)/2} + C e^{a(1+r)} \frac{z^{N+1}((N+1) - Nz)}{(1-z)^2}. \quad (4.7)$$

Both necklace inputs satisfy this estimate with $C = 1$.

4.2 Functional calculus, derivatives, and translation

Corollary 4.5 (Summable scalar operations). *If $F \in \mathcal{B}_{N,r}$ and $\|F\|_{N,r} = B < 1$, then $(1 + F)^{-1}$ and $\log(1 + F)$ converge in the Banach algebra. Their truncations satisfy*

$$\left\| (1 + F)^{-1} - \sum_{m=0}^M (-F)^m \right\|_{N,r} \leq \frac{B^{M+1}}{1 - B}, \quad (4.8)$$

$$\left\| \log(1 + F) - \sum_{m=1}^M \frac{(-1)^{m+1}}{m} F^m \right\|_{N,r} \leq \frac{B^{M+1}}{(M + 1)(1 - B)}. \quad (4.9)$$

More generally, a scalar analytic power series converges at F whenever its radius of convergence exceeds B . Evaluation commutes with these operations.

Proof. Submultiplicativity bounds the norm of each F^m by B^m . Sum the scalar absolute majorants. The evaluation maps have norm at most one and are multiplicative, so they commute with the norm limits. $\square$

Define the analytic-coordinate derivation by

$$D(e_l t^s) = -s e_l t^s. \quad (4.10)$$

It is not generally bounded on a single $\mathcal{B}_{N,r}$, but is bounded with a loss of margin.

Proposition 4.6 (Controlled loss of analytic margin). *For $0 \leq r' < r < N$ and $k \geq 0$,*

$$\|D^k F\|_{N,r'} \leq \frac{k!}{(r - r')^k} \|F\|_{N,r}. \quad (4.11)$$

Translation in the analytic coordinate, defined by $T_u(e_l t^s) = e^{-su} e_l t^s$, satisfies $\|T_u F\|_{N,r'} \leq \|F\|_{N,r}$ whenever $|u| \leq r - r'$. The label remains unchanged.

Proof. The weight ratio is $w_{N,r'}/w_{N,r} = e^{-s(r-r')}$. Use $s^k e^{-s(r-r')} \leq k!/(r - r')^k$, which follows from the k th term of the exponential series. The translation assertion uses $|e^{-su}| \leq e^{|s||u|}$. The Leibniz law for D follows from addition of the action indices in (3.5). $\square$

Thus the completion is a scale of analytic algebras rather than one Banach differential algebra with a bounded derivative. In particular, we have not silently identified differentiation in x with a shift or variation of the arithmetic label.

5 All-action inversion on arithmetic sheets

5.1 The exact core and the reversion equation

For large $y > 0$, the large positive solution of $e^{aX}/X = y$ is

$$X = -\frac{1}{a} W_{-1}\left(-\frac{a}{y}\right), \quad aX - \log X = \log y. \quad (5.1)$$

Here W_{-1} is the real branch of Lambert's function below -1 . Only the equation and the branch $X > 1/a$ are needed in the proofs; (5.1) is an exact convenient notation, not an asymptotic truncation. The same dominant core is already used in the repository [1].

Fix a label n and seek $x = X + \delta$ with $f_n(x) = e^{aX}/X$. After dividing the two core factors, this equation becomes

$$1 + E_n(X + \delta) = (1 + \delta/X)e^{-a\delta}. \quad (5.2)$$

Set

$$J_X(u) = \frac{1 - (1 + u/X)e^{-au}}{u}, \quad p = J_X(0) = a - \frac{1}{X}. \quad (5.3)$$

The removable value at $u = 0$ is understood. Introducing a coupling parameter z gives the exact implicit equation

$$\delta J_X(\delta) + z E_n(X + \delta) = 0. \quad (5.4)$$

We expand in z at zero and then set $z = 1$. The coupling order counts how many primitive sectors participate; it is not an action cutoff.

5.2 The algebraic reversion identity

Lemma 5.1 (Lagrange coefficient identity). *For $\delta = zF(\delta)$ with $\delta(0) = 0$, over any commutative $\mathbb{Q}$ -algebra, the unique formal solution satisfies*

$$[z^m]G(\delta) = \frac{1}{m}[u^{m-1}]G'(u)F(u)^m, \quad m \geq 1. \quad (5.5)$$

The coefficients exist without an invertibility assumption on $F(0)$.

Proof. The coefficient of z^m in δ is determined recursively by lower coefficients, proving existence and uniqueness. When $F(0)$ is an indeterminate unit, formal residue substitution $z = u/F(u)$ gives

$$[z^m]G(\delta) = \text{Res}_{u=0} G(u) \left(\frac{F(u)^m}{u^{m+1}} - \frac{F(u)^{m-1}F'(u)}{u^m} \right) du.$$

The residue of the derivative of $G(u)F(u)^m/u^m$ is zero. Rearranging this identity gives (5.5). For a fixed m , both sides are polynomial expressions with rational coefficients in finitely many coefficients of F and G . An identity proved after localizing the universal polynomial ring at its nonzero indeterminate $F(0)$ is therefore already an identity in that polynomial ring, and specializes to any commutative $\mathbb{Q}$ -algebra. This includes rings with idempotents and zero divisors. $\square$

Apply the lemma to $F(u) = -E_n(X + u)/J_X(u)$ and $G(u) = u$. Then

$$\delta_n(X) = \sum_{m \geq 1} \delta_{m,n}(X), \quad (5.6)$$

$$\delta_{m,n}(X) = \frac{(-1)^m}{m}[u^{m-1}] \left(\frac{E_n(X + u)}{J_X(u)} \right)^m. \quad (5.7)$$

At first this is a formal expression. Expanding the power of E_n reveals the arithmetic action structure:

$$\delta_n(X) = \sum_{m \geq 1} \sum_{d_1, \dots, d_m \geq 2} C_m(s, X) \left(\prod_{i=1}^m c_{d_i} \right) e_{\text{lcm}(d_1, \dots, d_m)}(n) e^{-sX}, \quad (5.8)$$

$$s = \lambda_{d_1} + \dots + \lambda_{d_m},$$

$$C_m(s, X) = \frac{(-1)^m}{m}[u^{m-1}]e^{-su}J_X(u)^{-m}. \quad (5.9)$$

The inner sum is over ordered tuples. For each exact s it is finite, including the possible m , by Lemma 3.1. The coefficients $C_m(s, X)$ are rational functions of X and a , with $s/a \in \mathbb{Q}$, whenever $p \neq 0$. Thus (5.8) is a formal arithmetic Hahn series with these rational functions as an additional coefficient field. No fractional divisibility predicate has appeared.

5.3 A quantitative convergence theorem

Theorem 5.2 (Absolute inverse transport with an explicit tail). *Let $a > 0$, $X \geq 4/a$, and $0 < \rho \leq 1/(8a)$. Fix n and any complex coefficients c_d on its divisors. Define $A_n(X)$ by (2.5) without using its final upper bound, and set*

$$\theta = \frac{2e^{a\rho} A_n(X)}{a\rho}. \quad (5.10)$$

If $\theta < 1$, equation (5.2) has a unique solution δ in $|\delta| < \rho$. It is the sum (5.6), and both that sum and its full ordered-tuple expansion (5.8) converge absolutely. For every $M \geq 0$,

$$\left| \delta - \sum_{m=1}^M \delta_{m,n}(X) \right| \leq \frac{\rho \theta^{M+1}}{(M+1)(1-\theta)}. \quad (5.11)$$

The same expression bounds the sum of the absolute values of all omitted tuple contributions.

Proof. First bound the denominator on the closed disk. Integrating the derivative of $(1 + u/X)e^{-au}$ gives

$$J_X(u) = \int_0^1 e^{-atu} \left(p + \frac{atu}{X} \right) dt. \quad (5.12)$$

Here $3a/4 \leq p < a$ and $\rho/X \leq 1/32$. Consequently,

$$|J_X(u) - p| \leq a(e^{1/8} - 1) + \frac{a}{32}e^{1/8} \leq \frac{a}{7} + \frac{a}{28} = \frac{5a}{28}.$$

The last inequality follows from $e^{1/8} \leq (1 - 1/8)^{-1} = 8/7$. It follows that $|J_X(u)| \geq 4a/7 > a/2$ throughout the disk. In particular, J_X has no zero there.

On $|u| = \rho$,

$$|E_n(X + u)| \leq e^{a\rho} A_n(X), \quad |uJ_X(u)| > a\rho/2.$$

For $|z| < 1/\theta$, Rouché's theorem applied to $uJ_X(u) + zE_n(X + u)$ gives exactly one zero in the disk, counted with multiplicity. It is therefore a simple zero. The local holomorphic implicit solutions patch, by uniqueness, into one holomorphic function $\delta(z)$ on $|z| < 1/\theta$ with $\delta(0) = 0$. At $z = 1$ it solves (5.2). Its Taylor coefficients are (5.7) by Lemma 5.1.

For absolute summation at the finer tuple level, the Cauchy coefficient bound in (5.9) is

$$|C_m(s, X)| \leq \frac{1}{m\rho^{m-1}} \left(\frac{2}{a} \right)^m e^{s\rho}. \quad (5.13)$$

For a fixed label only divisor tuples survive. Summing the absolute values of their contributions and using $\lambda_d < a$ gives

$$\sum_{\substack{d_1, \dots, d_m | n \\ d_i \geq 2}} |C_m(s, X)| \prod_i |c_{d_i}| e^{-sX} \leq \frac{\rho}{m} \left(\frac{2e^{a\rho} A_n(X)}{a\rho} \right)^m = \frac{\rho}{m} \theta^m.$$

Summing this bound from $m = M + 1$ to infinity proves (5.11), since $1/m \leq 1/(M + 1)$ on that range. It also proves absolute convergence before grouping actions, so all regroupings used in (5.8) are legitimate. $\square$

Remark 5.3 (The vanishing-input case). If $A_n(X) = 0$, then E_n is zero, the unique root in the disk is $\delta = 0$, and all coefficients and error bounds vanish. The proof using $1/\theta$ is read separately in this case.

Corollary 5.4 (Uniformity supplied by the Banach completion). *Suppose $\sum_{d \geq 2} |c_d|e^{-ad} < \infty$, and fix $0 < \rho \leq 1/(8a)$. For every sufficiently large integer N , inversion is uniformly convergent for all labels n and real core coordinates*

$$X \geq \max(N, n) - \rho, \quad X \geq 4/a. \quad (5.14)$$

An admissible common majorant in (5.11) is

$$\Theta_N = \frac{2}{a\rho} \|E\|_{N, 2\rho} < 1. \quad (5.15)$$

Proof. Choose $N > 2\rho$. For $|u| \leq \rho$ and a divisor $d \mid n$,

$$|c_d e^{-\lambda_d(X+u)}| \leq |c_d| e^{-\lambda_d(h_N(d)-2\rho)}.$$

The total absolute majorant on the circle is at most $\|E\|_{N, 2\rho}$. Repeat the proof of Theorem 5.2 with this majorant instead of $e^{a\rho} A_n(X)$, including the ordered-tuple estimate. The norm tends to zero by Theorem 4.3; hence $\Theta_N < 1$ eventually. $\square$

This corollary makes the connection between the global arithmetic completion and the local inverse theorem explicit. The inverse's coefficients depend rationally on X ; the assertion is a uniformly convergent family of such inverses, not an unannounced claim that variable- X coefficients belong to the original constant-coefficient Banach space.

5.4 An exact inverse on the range

Theorem 5.5 (Arithmetic range inverse). *Let c_d be real and satisfy the admission condition (4.4). Put*

$$y_n = \frac{e^{an}}{n} (1 + E_n(n)). \quad (5.16)$$

For all sufficiently large n , this is positive, and its large core inverse $X_n > 1/a$ satisfies $X_n - n \rightarrow 0$. The convergent series (5.8), evaluated with label n and core X_n , obeys

$$n = X_n + \sum_{m \geq 1} \delta_{m,n}(X_n). \quad (5.17)$$

For necklaces and primitive necklaces,

$$X_n - n = \mathcal{O}_a\left(ne^{-an/2}\right), \quad (5.18)$$

and the explicit bound (5.11) applies with $\theta \leq 2e^{a\rho} ne^{-aX_n/2}/(a\rho)$.

Proof. By Theorem 4.3 and diagonal evaluation, $|E_n(n)| \leq \|E\|_{n,0} \rightarrow 0$. In particular $1 + E_n(n) > 0$ for large n . Write $H(x) = ax - \log x$. Its derivative is $a - 1/x \geq a/2$ for $x \geq 2/a$, and

$$H(X_n) - H(n) = \log(1 + E_n(n)).$$

For large n the large core solution is also at least $2/a$, and the mean value theorem gives $|X_n - n| \leq (2/a) |\log(1 + E_n(n))| \rightarrow 0$. For the two counting inputs, (2.5) gives (5.18) after using $|\log(1 + t)| \leq 2|t|$ for $|t| \leq 1/2$.

Fix a permitted ρ . Eventually $|X_n - n| < \rho$ and the uniform inverse theorem applies. The number $\delta = n - X_n$ already satisfies (5.2), because $f_n(n) = y_n = e^{aX_n}/X_n$. Its size is less than ρ , so uniqueness identifies it with the convergent Lagrange series. The stated explicit bound follows from (2.5). $\square$

Remark 5.6 (What this does and does not invert). Equation (5.17) is an identity on the exact sequence range and a convergent local inverse on each specified sheet. It does not require choosing a global interpolation. It also does not by itself solve the problem of determining an unknown label from an arbitrary real target: the label occurs in the sheet data. On the exact range, (5.18) already shows that nearest-integer rounding of X_n recovers n eventually. A practical threshold problem can use candidate labels and exact comparisons, as in the repository; no new general threshold theorem is claimed here.

6 Coefficients, collisions, and the first post-cut sector

6.1 The first three homogeneous kernels

Expand $J_X(u) = p + j_1u + j_2u^2 + \dots$. Directly from (5.3),

$$j_1 = -\frac{a^2}{2} + \frac{a}{X}, \quad j_2 = \frac{a^3}{6} - \frac{a^2}{2X}. \quad (6.1)$$

The first kernels from (5.9) are

$$C_1(s, X) = -\frac{1}{p}, \quad (6.2)$$

$$C_2(s, X) = -\frac{s}{2p^2} - \frac{j_1}{p^3}, \quad (6.3)$$

$$C_3(s, X) = -\frac{s^2}{6p^3} - \frac{sj_1}{p^4} - \frac{2j_1^2}{p^5} + \frac{j_2}{p^4}. \quad (6.4)$$

For example, the second expression comes from the coefficient of u in $e^{-su}J_X(u)^{-2}/2$. The third follows from the coefficient of u^2 in the same expansion with exponent -3 , multiplied by $-1/3$. Equivalently,

$$\delta_{1,n} = -\frac{E_n(X)}{p}, \quad \delta_{2,n} = \frac{E_n(X)E'_n(X)}{p^2} - \frac{j_1E_n(X)^2}{p^3}. \quad (6.5)$$

This displays why differentiating the small sectors while holding their arithmetic indicators fixed is necessary.

6.2 The arithmetic content of action collisions

At action a , the only representation is $\lambda_2 + \lambda_2 = a$. At action $4a/3$, the unordered two-term representations are $(2, 6)$ and $(3, 3)$. At action $3a/2$, they are $(3, 6)$ and $(4, 4)$, and there is also the three-term representation $(2, 2, 2)$. There are no one-term representations at these actions. The completeness of these short lists can be checked by solving the corresponding two-unit-fraction equations, or by the finite algorithm in Section 9.

For $c_d = \varphi(d)$, the coefficients multiplying the indicated exponential monomial are

$$\begin{aligned} [e^{-aX}]\delta_n &= C_2(a, X)e_2(n), \\ [e^{-4aX/3}]\delta_n &= C_2(4a/3, X)(4e_3(n) + 4e_6(n)), \\ [e^{-3aX/2}]\delta_n &= C_2(3a/2, X)(8e_6(n) + 4e_4(n)) + C_3(3a/2, X)e_2(n). \end{aligned} \quad (6.6)$$

The bracket notation here means an exact action coefficient in the formal series, not extraction from a one-variable ordinary power series. For $c_d = \mu(d)$ the corresponding expressions are

$$\begin{aligned} [e^{-aX}] \delta_n &= C_2(a, X) e_2(n), \\ [e^{-4aX/3}] \delta_n &= C_2(4a/3, X) (e_3(n) - 2e_6(n)), \\ [e^{-3aX/2}] \delta_n &= -2C_2(3a/2, X) e_6(n) - C_3(3a/2, X) e_2(n). \end{aligned} \quad (6.7)$$

For instance the two ordered tuples $(2, 6)$ and $(6, 2)$ contribute $2c_2c_6e_6$, whereas $(3, 3)$ contributes $c_3^2e_3$. Keeping only the total action would lose this distinction.

6.3 Cancellation at the accumulation boundary

Substitute $s = a$ into (6.3). A cancellation gives the exact identity

$$C_2(a, X) = -\frac{a}{2X(a - 1/X)^3}. \quad (6.8)$$

Its limit as $X \rightarrow \infty$ is zero. Thus an action at the first nonlinear boundary is present, but its amplitude is one algebraic order smaller than a naive nonresonant estimate suggests.

Theorem 6.1 (First nonlinear sector beyond the primitive cut). *Fix $a = \log q > 0$ and either $c_d = \varphi(d)$ or $c_d = \mu(d)$. Fix $0 < \rho \leq 1/(8a)$. Uniformly for real X with $|X - n| \leq \rho$ as $n \rightarrow \infty$, the inverse displacement of Theorem 5.2 satisfies*

$$\delta_n(X) + \frac{E_n(X)}{a - 1/X} = -\frac{a e_2(n)}{2X(a - 1/X)^3} e^{-aX} + \mathcal{O}_a(n^2 e^{-7aX/6}). \quad (6.9)$$

In particular, on even labels,

$$\delta_n(X) + \frac{E_n(X)}{a - 1/X} \sim -\frac{a}{2X(a - 1/X)^3} e^{-aX}. \quad (6.10)$$

The same statements hold at the exact range core $X = X_n$.

Proof. The first homogeneous term is exactly $-E_n(X)/p$, so it is removed in the left side. The unique quadratic tuple with action a is $(2, 2)$. In both counting cases $c_2^2 = 1$, and its label is e_2 . Its coefficient is (6.8).

Every other quadratic tuple has at least one index at least three, so its action is at least $\lambda_2 + \lambda_3 = 7a/6$. The kernel $C_2(s, X)$ is uniformly bounded for $s \in [a, 2a]$ and large X . The sum of absolute products of the divisor coefficients is at most n^2 , by (2.4). The remaining quadratic terms are therefore $\mathcal{O}_a(n^2 e^{-7aX/6})$.

For the remaining estimates, use the fixed disk radius $1/(8a)$ in Theorem 5.2; uniqueness identifies its root with the root in any smaller permitted disk for large n . This makes the constants depend only on a . For $m \geq 3$, its absolute tail is $\mathcal{O}_a(A_n(X)^3) = \mathcal{O}_a(n^3 e^{-3aX/2})$, uniformly in the stated range, since $\theta \rightarrow 0$. This is absorbed by the preceding bound: their ratio is $ne^{-aX/3} \rightarrow 0$. This proves (6.9). On even labels the ratio of the error to the absolute leading term is $\mathcal{O}_a(Xn^2 e^{-aX/6}) \rightarrow 0$, giving (6.10). The range assertion follows from (5.18). $\square$

Remark 6.2 (Why the full primitive block must be retained). The subtracted term $E_n(X)/p$ includes every divisor of n . For necklaces, if a divisor cutoff omits $d = n$, it omits the positive term $\varphi(n)e^{-aX+aX/n}$. When $X = n + \mathcal{O}(1)$, this term is at least a positive constant times e^{-aX} , already larger than the $X^{-1}e^{-aX}$ effect in (6.9). The theorem cannot be recovered by subtracting a fixed finite primitive prefix and then reading off the quadratic sector. This is exactly where summation across the accumulating block becomes mathematically necessary.

7 An all-order resonance law

The cancellation in (6.8) is not isolated. We now transform the coefficient kernel so that its entire pure-exponential limit, and the first correction caused by the factor x^{-1} , are explicit.

For complex α and $m \geq 0$, let $\binom{\alpha}{m} = \alpha(\alpha-1)\cdots(\alpha-m+1)/m!$. Define the analytic germ

$$B(w) = \frac{(1+w)\log(1+w)}{w}, \quad B(0) = 1, \quad \varepsilon = \frac{1}{aX}. \quad (7.1)$$

The logarithm is the germ at $w = 0$.

Theorem 7.1 (Exact transformed kernel and curvature lifting). *For every $m \geq 1$, s , and X with $p \neq 0$, put $\alpha = s/a$. The kernel (5.9) has the exact coefficient representation*

$$C_m(s, X) = -\frac{1}{am} [w^{m-1}] (1+w)^{\alpha-1} (1-\varepsilon B(w))^{-m}. \quad (7.2)$$

For fixed m, α , as $X \rightarrow \infty$,

$$C_m(s, X) = -\frac{1}{am} \binom{\alpha-1}{m-1} - \frac{1}{a^2 X} \frac{d}{d\alpha} \binom{\alpha}{m} + \mathcal{O}_{a,m,\alpha}(X^{-2}). \quad (7.3)$$

At every integer k with $1 \leq k < m$, the first term vanishes and

$$C_m(ka, X) = \frac{(-1)^{m-k}}{a^2 X m \binom{m-1}{k}} + \mathcal{O}_{a,m,k}(X^{-2}). \quad (7.4)$$

Whenever the action ka has an m -tuple representation in the arithmetic support, this is its universal homogeneous kernel. The arithmetic sum of tuple amplitudes can, independently, vanish.

Proof. It is useful to derive the identity for a finite collection of sector amplitudes $b_d = c_d e_d(n) e^{-\lambda_d X}$, treated as independent commuting quantities. In (5.2) with coupling z , put $v = e^{-a\delta}$ and $w = v - 1$. Since $\delta = -a^{-1} \log v$, the equation becomes

$$v(1 - \varepsilon \log v) = 1 + z \sum_d b_d v^{\lambda_d/a}.$$

Equivalently,

$$w(1 - \varepsilon B(w)) = z \sum_d b_d (1+w)^{\lambda_d/a}. \quad (7.5)$$

Use Lemma 5.1 on this equation with $G(w) = -a^{-1} \log(1+w)$. The contribution of an ordered tuple of length m is precisely the right side of (7.2), since $G'(w) = -1/(a(1+w))$ and the tuple's exponents sum to α . The identity is a universal coefficient identity, so it also proves equality with (5.9). Only finitely many Taylor coefficients are involved at fixed m .

Expand the last factor in (7.2) at $\varepsilon = 0$:

$$(1 - \varepsilon B(w))^{-m} = 1 + m\varepsilon B(w) + \mathcal{O}(\varepsilon^2).$$

The constant term in ε gives the first term of (7.3). For the linear term, the definition of B gives

$$\begin{aligned} [w^{m-1}] (1+w)^{\alpha-1} B(w) &= [w^m] (1+w)^\alpha \log(1+w) \\ &= \frac{d}{d\alpha} \binom{\alpha}{m}. \end{aligned}$$

Since $\varepsilon/a = 1/(a^2X)$, this proves (7.3). The remainder estimate is valid for fixed m, α : the relevant finite coefficient extraction is analytic in ε near zero.

At $\alpha = k \in \{1, \dots, m-1\}$, $\binom{k-1}{m-1} = 0$. Differentiating the polynomial $\binom{\alpha}{m}$ at its simple root k gives

$$\left. \frac{d}{d\alpha} \binom{\alpha}{m} \right|_{\alpha=k} = \frac{k!(-1)^{m-1-k}(m-1-k)!}{m!} = \frac{(-1)^{m-1-k}}{m \binom{m-1}{k}}.$$

Substitution proves (7.4). $\square$

For the necklace action set, an m -tuple has $m/2 \leq \alpha < m$. Thus only integers in this interval can occur as actual resonant actions, even though the kernel identity holds for all k in the theorem. For $m = 2$, $k = 1$, formula (7.4) reads $-1/(2a^2X) + \mathcal{O}(X^{-2})$, in agreement with the exact expression (6.8).

7.1 The pure-exponential comparison

If the factor x^{-1} is removed from the forward core, then J_X is replaced by $J_\infty(u) = (1 - e^{-au})/u$ and the kernel is exactly

$$C_m^{\text{exp}}(s) = -\frac{1}{am} \binom{s/a - 1}{m-1}. \quad (7.6)$$

This follows either by putting $\varepsilon = 0$ in (7.5), or by the same change of coordinate for the pure-exponential equation. The integer zeros in (7.6) are therefore exact cancellations of the entire homogeneous kernel. The logarithmic term in the phase $ax - \log x$ removes these zeros at order X^{-1} . This is the sense in which curvature lifts the resonance.

7.2 General power-law factors

Consider instead

$$f_n^{(b)}(x) = e^{ax} x^{-b} (1 + E_n(x)), \quad b \in \mathbb{R} \text{ fixed}, \quad (7.7)$$

near a large positive X , with the local analytic branch of x^{-b} . Replace J_X by

$$J_X^{(b)}(u) = \frac{1 - (1 + u/X)^b e^{-au}}{u}, \quad J_X^{(b)}(0) = a - b/X. \quad (7.8)$$

For fixed b this is uniformly nonzero on a sufficiently small fixed disk for all large X , because it converges there to J_∞ . Thus the same Rouché and coefficient arguments provide an inverse series and a geometric tail, with the lower bound for $J_X^{(b)}$ substituted in place of $a/2$.

Corollary 7.2 (Universal first curvature coefficient). *Let $C_m^{(b)}(s, X)$ be defined by (5.9) with $J_X^{(b)}$. For fixed $a, b, m, \alpha = s/a$,*

$$C_m^{(b)}(s, X) = -\frac{1}{am} \binom{\alpha - 1}{m-1} - \frac{b}{a^2X} \frac{d}{d\alpha} \binom{\alpha}{m} + \mathcal{O}(X^{-2}). \quad (7.9)$$

At an integer resonance $1 \leq k < m$ this gives

$$C_m^{(b)}(ka, X) = \frac{(-1)^{m-k}b}{a^2X m \binom{m-1}{k}} + \mathcal{O}(X^{-2}). \quad (7.10)$$

Proof. The change $v = 1 + w = e^{-a\delta}$ now gives

$$(1 + w)(1 - \varepsilon \log(1 + w))^b = 1 + z \sum_d b_d (1 + w)^{\lambda_d/a}.$$

After subtracting one and dividing by w , the factor multiplying w is

$$K_b(w, \varepsilon) = \frac{(1 + w)(1 - \varepsilon \log(1 + w))^b - 1}{w} = 1 - b\varepsilon B(w) + \mathcal{O}(\varepsilon^2).$$

All expressions are analytic germs, including at $w = 0$. Lagrange extraction now contains K_b^{-m} , whose expansion is $1 + mb\varepsilon B(w) + \mathcal{O}(\varepsilon^2)$. Repeat the coefficient calculation in Theorem 7.1. The simple-root derivative formula then gives (7.10). $\square$

The bounds in (7.3) and (7.9) are for fixed homogeneous order. They are not uniform when m grows with X . The joint large-order problem is genuinely additional, as noted in Section 11.

8 Obstructions to stronger interpretations

The limitations below are structural statements with proofs, not missing steps concealed inside the inversion theorem.

8.1 Indicators cannot live faithfully in a field

Proposition 8.1 (Idempotent obstruction). *No unital ring embedding of the arithmetic coefficient ring $\mathcal{A}$ into a field exists. Every derivation on a commutative ring containing the indicators annihilates each e_d . Consequently the shift $A(n) \mapsto A(n+1)$ cannot be the exponential of a derivation that preserves the indicator ring.*

Proof. The indicator e_2 is a nonzero idempotent different from one, and $e_2(1 - e_2) = 0$ with both factors nonzero. A field has no such pair, so an injective ring map is impossible.

For any idempotent e , the Leibniz rule gives $D(e) = D(e^2) = 2eD(e)$, whence $(1 - 2e)D(e) = 0$. But $(1 - 2e)^2 = 1$, so $D(e) = 0$. Thus every iterate of D annihilates e_d , and any meaningful exponential of D fixes e_d . The integer shift sends e_2 to $1 - e_2$, so it cannot be such an exponential. $\square$

The arithmetic ring therefore is not an ordinary transseries field with an enlarged set of constant symbols. A difference operator acting on labels can be studied separately, but it is not supplied by analytic translation in x . This is why the frozen-label convention is necessary rather than cosmetic.

8.2 A compact profinite parameter does not repair summation

Let $\widehat{\mathbb{Z}}$ denote the profinite completion of the integers. Each single indicator e_d extends continuously to it, because it is the characteristic function of the clopen set $d\widehat{\mathbb{Z}}$. It is tempting to sum the primitive block at fixed x and regard the result as a continuous function of a profinite label. The following proposition shows that this fails for both counting inputs.

Proposition 8.2 (Fixed-coordinate profinite obstruction). *Fix a real $x > 0$. Neither of the functions $n \mapsto E_n(x)$ for $c_d = \varphi(d)$ or $c_d = \mu(d)$ is the restriction to positive integers of a continuous complex-valued function on $\widehat{\mathbb{Z}}$.*

Proof. Put $b = ax > 0$. In the totient case, $\lambda_d < a$ and positivity imply

$$E_n(x) \geq e^{-b} \sum_{d|n, d \geq 2} \varphi(d) = e^{-b}(n-1).$$

Thus the restriction is unbounded, whereas every continuous function on the compact space $\widehat{\mathbb{Z}}$ is bounded.

For the Möbius case, including the $d = 1$ term gives

$$1 + E_n(x) = e^{-b} \sum_{d|n} \mu(d) e^{b/d}.$$

For $n > 1$, expand the exponential and interchange with the finite divisor sum. The constant coefficient vanishes, and

$$\sum_{d|n} \mu(d) e^{b/d} = \sum_{k \geq 1} \frac{b^k}{k!} \prod_{p|n} (1 - p^{-k}). \quad (8.1)$$

Every product is between zero and one, so $1 + E_n(x) \leq 1 - e^{-b}$, or $E_n(x) \leq -e^{-b}$, for all $n > 1$. On the other hand, $E_1(x) = 0$. The positive integers $n_j = 1 + j!$ converge to 1 in the profinite topology: every fixed modulus divides $j!$ eventually. The values $E_{n_j}(x)$ stay separated from zero, contradicting continuity at 1. $\square$

The weighted sheet domain avoids this contradiction by requiring the analytic coordinate to grow with the label. It is not a product of a fixed analytic domain with all of $\widehat{\mathbb{Z}}$. Other weighted or partially defined profinite constructions remain possible, but they must respect the obstruction just proved.

8.3 Bounded-action extraction versus homogeneous truncation

Theorem 5.2 truncates at a homogeneous order M . Each retained term sums the entire primitive divisor block, and the omitted terms carry at least $M + 1$ primitive factors. This is different from listing all actions below a fixed numerical cutoff. Before action a alone there are infinitely many primitive actions in the universal series. No theorem above converts that infinite list into a finite grid.

At a fixed label n there are only finitely many primitive divisors, so a computation is finite. Uniformity over all n is supplied by the weighted norm, not by silently interchanging a label-dependent finite sum with a globally finite action list. Equally, summability in this article means ordinary absolute convergence of the specified families; no Borel summation, alien derivative, or Stokes automorphism is asserted.

9 Finite algorithms and reproducible audits

9.1 Exact membership and a fixed-action coefficient algorithm

Proposition 9.1 (A finite action search). *For a given rational $\alpha \geq 0$, it is decidable by a finite exact search whether $\alpha a \in S$, and all generator representations can be enumerated. Hence every exact action coefficient in (5.8) is computable from a finite list of the input coefficients c_d .*

Proof. For $\alpha = 0$ take the empty tuple. Otherwise a representation of length m must have $m \leq \lfloor 2\alpha \rfloor$ and obey

$$\frac{1}{d_1} + \cdots + \frac{1}{d_m} = m - \alpha > 0. \quad (9.1)$$

Search only sorted denominators. At a recursive step with k terms remaining, target sum $t > 0$, and smallest permitted denominator L , the next denominator lies in the finite interval

$$\max\left(L, \left\lceil \frac{1}{t} \right\rceil\right) \leq d \leq \left\lfloor \frac{k}{t} \right\rfloor. \quad (9.2)$$

The lower bound follows from $1/d \leq t$. The upper bound follows because the remaining sorted denominators are at least d , so the sum of the k reciprocals is at most k/d . Recurse with target $t - 1/d$, $k - 1$ terms, and lower bound d . Nonpositive targets are rejected unless no terms remain and the target is zero. For $k = 1$ the sole possible denominator is $1/t$, which must be an integer at least L . The depth and every branching interval are finite.

Restore the ordered tuples by permutation. Equivalently, a sorted tuple with multiplicities r_d receives the factor $m! / \prod_d r_d!$. The action coefficient follows by multiplying the corresponding c_d , using the lcm of the denominators for the label, and applying (5.9). No bound on the running time polynomial in the bit length of α is claimed. $\square$

9.2 Efficient homogeneous jets for a known label

Enumerating all tuples is unnecessary for numerical inversion on one sheet. Given the divisors of n , form the weighted moments

$$W_k = \sum_{d|n, d \geq 2} c_d \lambda_d^k e^{-\lambda_d X}, \quad 0 \leq k < M. \quad (9.3)$$

The Taylor coefficients required by (5.7) are

$$[u^k]E_n(X + u) = \frac{(-1)^k}{k!} W_k, \quad (9.4)$$

$$[u^k]J_X(u) = -\frac{(-a)^{k+1}}{(k+1)!} - \frac{(-a)^k}{X k!}. \quad (9.5)$$

Divide the two jets modulo u^M , obtaining $F(u) = E_n(X + u)/J_X(u)$. Successively multiply by F modulo u^M and read $(-1)^m [u^{m-1}]F(u)^m/m$ for $m = 1, \dots, M$.

With naive polynomial arithmetic, the moment construction and these jet operations take $\mathcal{O}(M\tau(n) + M^3)$ scalar arithmetic operations once the divisors and exponentials are available. This is not a bit-complexity or integer-factorization bound. The supplied implementation uses trial division to enumerate divisors, and can be replaced by a faster divisor provider without changing the coefficient calculation. Large precision is important when checking an exponentially small residual by subtracting two nearly equal inverse values.

For a genuinely machine-certified numerical enclosure, every input bound and polynomial operation must use outward-rounded intervals. The analytic inequality (5.11) is rigorous; the recorded floating-point evaluations of it are diagnostic values, not interval certificates.

9.3 What the accompanying program checks

The file `verification/verify.py` is self-contained except for `mpmath`. Its exact portion uses Python integer arithmetic and `fractions.Fraction`. It performs 964 successful exact assertions: it checks finite lcm-convolution identities, divisor identities up to $n = 300$, substitution

of the rational formal inverse through coupling order seven, the integer-root derivative formula through $m = 14$, four specified action-collision lists, and 80 finite- X identities comparing the two coefficient coordinates in (5.9) and (7.2).

The numerical portion uses 400 decimal digits. Exact integer necklace or primitive-necklace counts provide the targets, not floating-point forward approximations. For twelve combinations of counting type, q , and n , it computes the large Lambert core X , compares the first four homogeneous inverse sums with the exact displacement $n - X$, and checks all 48 instances of (5.11). It also records ratios testing (6.10). The generated CSV and JSON files are included so that the reported checks can be rerun.

Table 1: Binary-necklace range inversion. Actual absolute error of the first M homogeneous terms, and the evaluated bound from (5.11). These are high-precision diagnostics, not outward-rounded intervals. Values are generated by the supplied program.

n	M	Absolute error	Analytic-bound value
20	1	1.708×10^{-7}	3.853×10^{-5}
20	2	4.958×10^{-11}	5.255×10^{-7}
20	4	2.195×10^{-18}	1.32×10^{-10}
60	1	1.671×10^{-20}	2.581×10^{-17}
60	2	4.953×10^{-29}	2.912×10^{-25}
60	4	4.775×10^{-48}	5.001×10^{-41}
120	1	6.766×10^{-39}	2.23×10^{-35}
120	2	3.961×10^{-56}	2.338×10^{-52}
120	4	3.341×10^{-93}	3.469×10^{-86}

For example, at $q = 2, n = 120$ the ratio of the left side of (6.10) to its proposed leading term is approximately 1.0000561453 for necklaces and 1.0000280316 for primitive necklaces. At $q = 3, n = 210$ the deviations of these ratios from one are approximately 3.14×10^{-15} and 1.57×10^{-15} , respectively. These observations agree with the proved asymptotic estimate. At smaller labels the ratio can be substantially different from one; the bound and proof, not a visual numerical trend, justify the asymptotic statement.

The audits are finite tests, not substitutes for the general proofs. In particular they do not establish the Banach completeness theorem, the infinite support argument, or the uniform asymptotic statements. No Lean kernel check of the new theorems was performed.

10 Consequences and interpretation

10.1 A usable answer at an infinite action boundary

The construction supplies two complementary kinds of finiteness. For algebraic equality at one action, the support lemma makes the coefficient extraction finite. For evaluation across an entire accumulating block, the action-label norm supplies absolute convergence. Neither mechanism can replace the other: a well-ordered support does not by itself make its evaluation summable, and pointwise convergence on one sheet does not by itself provide a faithful coefficient calculus.

The inverse theorem then uses a third organization, homogeneous degree in the primitive input. Its analytic majorant remains geometric even though a bounded-action prefix of the universal input is infinite. This separation of exact action extraction, arithmetic labeling, and analytic homogeneous summation is the main structural contribution.

10.2 A precise beyond-all-orders effect

The leading range displacement $-E_n(X)/p$ already contains infinitely many universal action sectors. Once that entire block is retained, (6.9) detects a specific smaller effect at its accumulation boundary. Its coefficient is neither a freely chosen exponentially small constant nor an interpolation-dependent Stokes parameter. It is forced by the exact inverse equation and the identity $e_2^2 = e_2$. The factor X^{-1} is forced by the slowly varying part of the core.

More generally, (7.10) separates two possible reasons an action coefficient can vanish. The universal analytic kernel may vanish in the pure-exponential limit, or the arithmetic sum over its lcm-labeled tuples may cancel. Curvature resolves the first mechanism at order X^{-1} when $b \neq 0$; it does not automatically resolve the second. This distinction remains meaningful at action collisions such as those in (6.6) and (6.7).

10.3 What remains outside the theorem

The article does not construct the largest possible arithmetic transseries ring, prove a general closure theorem for arbitrary compositions, or define divisibility for noninteger labels. The norm criterion is exact for the stated elementary block, not a classification of every conditional or regularized summation method. The inverse theorem is local in the analytic coordinate and is uniform only in its stated large-core domains. A critical point near $X = 1/a$ is excluded. The curvature formulas are fixed-order expansions, and the numerical checks use no interval arithmetic. These boundaries leave several substantive research problems.

11 Further research questions

Research question 1 (Joint action–label admissibility beyond one block). For a general array $c_{s,l}$, characterize those formal arithmetic Hahn series that admit a faithful holomorphic sheet realization with uniform nonlinear inversion. Can an intrinsic condition, independent of a chosen family $h_N(l)$, recover the useful part of the weighted completion without forcing absolute ℓ^1 summability?

A useful first benchmark is to compare the present norm with a norm formed after summing absolute coefficients on each divisor fiber. The latter can be substantially smaller, but its multiplicative and faithfulness properties require separate analysis. Conditional summation should not be conflated with the elementary-block criterion already proved.

Research question 2 (Difference operations on arithmetic labels). Construct a compatible arithmetic difference calculus generated by congruence indicators $e_{d,r}(n) = \mathbf{1}_{n \equiv r \pmod{d}}$, in which integer shifts act naturally. Determine which joint action–congruence weights make multiplication and shift uniformly continuous.

Products of congruence indicators are either zero or another congruence indicator, according to the Chinese remainder compatibility condition. This suggests an extension of the lcm rule, but Proposition 8.1 still prevents identifying label shifts with the analytic exponential of a derivation. A successful theory should expose both operations rather than hide this distinction.

Research question 3 (Uniform large homogeneous order). Determine the joint asymptotics of $C_m(ka, X)$ when m and k grow with X . At what scale does the first curvature term in (7.4) cease to dominate, and what rescaled profile replaces it? The exact representation (7.2) supplies a concrete starting point for a two-parameter saddle analysis.

The present theorem controls fixed m, k only. Expanding in $1/X$ and then allowing m to grow is not justified by its proof. A uniform result could connect geometric homogeneous convergence to sharp optimal truncation within individual action families.

Research question 4 (Arithmetic cancellation at resonant actions). Classify the labels and coefficient sequences for which

$$\sum_{\lambda_{d_1} + \dots + \lambda_{d_m} = ka} \left(\prod_i c_{d_i} \right) e_{\text{lcm}(d_1, \dots, d_m)}$$

vanishes identically, or on specified arithmetic subsequences. How do totient positivity and Möbius signs alter the first surviving curvature-lifted term?

The finite action algorithm makes each fixed case decidable. A structural classification would require more than checking initial coefficients, especially where several lcm labels survive at the same action.

Research question 5 (More complicated accumulation sets). Extend the action–label completion to positive generator sets with several finite accumulation levels or higher order types. Identify conditions that separately guarantee well-ordering, finite exact-action decompositions, and convergence of the label-weighted norm.

The positive lower bound $a/2$ bounds homogeneous length in the present proof. Removing it can destroy this argument even when individual labels leave only finitely many input terms. A replacement theorem would need to control both arbitrarily long products and any action accumulation at zero.

Research question 6 (Optimal expanding domains). For c_d close to the threshold $|c_d| \asymp e^{ad}$, determine the smallest label-dependent analytic half-planes on which inverse families remain uniformly summable. Can the norm be sharpened sufficiently to describe logarithmic or iterated-logarithmic boundary corrections?

For example, elementary inputs with $|c_d| = e^{ad}/d^\gamma$ are admitted exactly for $\gamma > 1$ in the present norm. At or below that boundary, every fixed sheet still exists; the issue is the geometry and summability of the family, not the existence of the finite divisor sum.

Research question 7 (Certified numerical realization). Implement the jet algorithm with outward-rounded intervals, including the large Lambert core and an explicit divisor-coefficient majorant. Can the majorant be sharpened by action-aware estimates to provide near-optimal certified errors without enumerating all tuples?

The current Cauchy bound is explicit but can be conservative, as Table 1 shows. A proof-producing implementation should separate the exact combinatorial count, interval evaluation of the core, and propagation of the inverse tail.

Research question 8 (Formal verification of the arithmetic completion). Formalize the action-support theorem, lcm-coordinate ring, weight inequality, and homogeneous coefficient identities in Lean. Then connect a finite checker to the mathematical semantics of the jet algorithm and to the analytic tail theorem.

Appendix B gives a dependency order and the limited existing repository interfaces that can be reused. The purpose is to verify the new construction, not to relabel an unverified analytic argument as an already formalized theorem.

Research question 9 (Critical cores with arithmetic sector accumulation). Study inverse sheets when the core approaches its critical slope $a - 1/X = 0$ while an arithmetic primitive block remains summable. Is there a uniform fold coordinate compatible with the full lcm-labeled block, and how does it interact with the action accumulation cut?

The geometric disk proof here deliberately assumes a lower slope bound. Existing repository fold analyses concern other models. Combining such coordinates with the present arithmetic summation would be a separate extension, not a consequence of the large- X theorem.

12 Conclusion

An accumulating action support need not prevent exact symbolic multiplication, and arithmetic indicators need not be replaced by trigonometric surrogates to obtain a summable analytic theory. The lcm-coordinate ring, a well-ordered positive action monoid, and a weight that couples action to the earliest possible arithmetic label together provide a workable alternative. The resulting inverse expansion is absolutely convergent on specified sheets and exact on the counting sequence's range.

The first post-cut inverse sector illustrates why this framework is more than a reformulation of finite divisor cutoffs: the whole primitive block must first be retained, after which a curvature-induced $X^{-1}e^{-aX}$ term becomes visible. The all-order kernel identity explains its resonance mechanism and its extension to a general power-law core. These are precise research results within an explicit scope; their broader novelty, optimized numerical realization, and formal verification remain tasks for further work.

A Repository provenance and scope ledger

The repository was inspected through its GitHub connector, with a subsequent pinned read of the exact open-question passage. The snapshot used for this article is

db68f0853c3c69cd930caedab6f6ad9addb11eaf.

Its parent was 04473354a0f3edff2d6c365caf26170ca6b88735. The snapshot's final commit concerns unrelated ordinal reports; the transseries source blob recorded below is unchanged by that commit. The article does not depend on any unrelated claims in the repository.

Primary mathematical source. Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex

The source blob is 17647298fa31d61c061d556a3b1e451a228ae20e. The necklace chapter and its finite cutoffs were read, along with the synthesis and research directions. The explicit arithmetic-algebra question was reread at the pinned snapshot, source lines 4955–4978.

Comparison with existing arrivals. Analysis/Transseries/docs/series-and-transseries/README.md

Its entry for `Action_Accumulation_Nonlinear_Inversion` states that that article does not retain divisibility indicators. The catalogue was used to avoid duplicating an analytic action-measure result. No complete audit of all repository research reports was attempted.

Existing Lean source inspected. `Analysis/Transseries/Lean/Transseries/TransseriesWellBased.lean`

Its blob is `06d5ea1663573d96d5712636347d44764b40f800`. The actual declarations, rather than only a catalogue description, were read for the formalization comparison below.

The package also includes `source_notes/provenance.json` and `source_notes/scope.md`. These record what was consulted and what was not claimed. No copy of the entire repository or its large source volumes is required to compile this article.

B Lean interfaces and a proposed verification route

The inspected module places its declarations in namespace `Fabius`, although its module path is now under `Transseries`. It supplies `dickson_isPWO`, `dickson_antichain_finite`, and `dickson_isPWO_pi`; the Neumann interfaces include `neumann_isPWO` and `neumann_finite_factorizations`, together with additive versions generated by `to_additive` and explicit order-dual wrappers [3].

The source describes these as bridges to Mathlib’s partially well-ordered-set and finite-antidiagonal results. Their presence is useful, but it is important not to overstate it. They do not instantiate the lcm-labeled Banach algebra, prove its norm inequality, verify the new analytic inverse theorem, or establish the curvature formula. No such formalization is delivered with this article.

A proposed dependency order is as follows.

1. Define the rational normalized actions $1 - 1/d$ for $d \geq 2$. Use the finite-length bound and the available Dickson interface to prove well-ordering and finite exact-action representations. Keep the embedding into real actions $a\alpha$ separate from the rational combinatorial support.
2. Define divisor-coordinate arithmetic functions and prove $e_i e_j = e_{\text{lcm}(i,j)}$. Establish the finite lcm convolution, its associativity, and the divisor-zeta equivalence. This stage needs no analysis or asymptotics.
3. Construct the coefficient ring on the action monoid, proving finiteness before defining each convolution. Verify the universal polynomial identities underlying finite Lagrange jets over a commutative rational algebra.
4. Introduce the weighted summable-family space. Prove the single weight inequality from Theorem 4.1, then use it to build the normed algebra and its evaluation homomorphisms. Establish the admission equivalence by comparison of positive series.
5. Formalize the analytic disk bounds, root existence and uniqueness, and Cauchy coefficient estimates. Finally connect the finite jet checker to this analytic theorem. The finite program should not be trusted as a source of unproved universal identities.

The exact derivative-of-binomial identity in Theorem 7.1 and the finite action search are plausible early verification targets. The uniform holomorphic-family construction is a substantially larger step. This is an implementation roadmap, not a report of a Lean build.

C Reproducibility and deliverable contents

The source is a standalone LaTeX article. From its directory, run

```
python verification/verify.py
pdflatex -interaction=nonstopmode -halt-on-error arithmetic_transseries.tex
pdflatex -interaction=nonstopmode -halt-on-error arithmetic_transseries.tex
pdflatex -interaction=nonstopmode -halt-on-error arithmetic_transseries.tex
```

The equivalent convenience command is `sh build.sh`. A recent Python with `mpmath` and a standard TeX installation with the packages used in the preamble are required. The generated verification table is embedded in this standalone source; a matching table fragment is also shipped. The article can therefore be compiled without rerunning the numerical script.

The verification program records its working precision and separates exact checks from numerical diagnostics. The counts in this article refer to the program included in this package, not to any pre-existing repository audit. The PDF was compiled from the delivered source and its rendered pages were inspected. No font files, third-party source volumes, or build-cache files are included in the archive.

References

- [1] Vladimir Reshetnikov, *ProveIt: Combinatorial Transseries and Their Inverses*, repository manuscript, pinned snapshot of 29 September 2026. Primary path, source blob, and inspected passages are specified in Appendix A. <https://github.com/VladimirReshetnikov/ProveIt/tree/db68f0853c3c69cd930caedab6f6ad9addb11eaf/Analysis/Transseries>.
- [2] Vladimir Reshetnikov, *Series and transseries*, ProveIt catalogue, including the arrivals of 29 September 2026 and the scope description of *Action Accumulation and Nonlinear Inversion*. Repository and snapshot as in [1]; path `Analysis/Transseries/docs/series-and-transseries/README.md`.
- [3] Vladimir Reshetnikov, *TransseriesWellBased.lean*, ProveIt Lean source, snapshot as in [1]; path and blob in Appendix A.
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